

Executive Summary

Project Title: Machine Learning–Based Late Delivery Risk Prediction in Global Supply Chain Operations

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Overview

Late deliveries in supply chain operations lead to increased costs, customer dissatisfaction, and operational inefficiencies. This project focuses on developing a predictive model to identify the risk of late deliveries before shipment, enabling proactive decision-making.

Objective

The primary objective of this project is to build a machine learning model that can predict whether an order is likely to be delivered late, allowing logistics teams to take preventive actions.

Approach

The project uses historical order data containing information on shipping timelines, product details, and customer transactions. The data was preprocessed and enhanced using feature engineering techniques such as:

- Delay Gap (difference between actual and scheduled shipping time)
- Order Complexity (based on quantity and product price)

Two machine learning models were developed:

- Logistic Regression (baseline model)
- Random Forest (final model)

Key Results

- Random Forest model achieved an accuracy of approximately **97.8%**
- ROC-AUC score of **0.998**, indicating excellent prediction capability
- The model successfully classifies orders into:
 - Low Risk
 - Medium Risk
 - High Risk

Key Insights

The most significant factors contributing to delivery delays include:

- Delay Gap between actual and scheduled shipping
- Shipping duration
- Shipping mode

Business Impact

The implementation of this model can help organizations:

- Identify high-risk orders in advance
- Optimize delivery planning and resource allocation
- Reduce penalties and operational costs
- Improve customer satisfaction

Conclusion

This project demonstrates how machine learning can transform supply chain management from reactive problem-solving to proactive risk management. The developed model provides a reliable and efficient tool for predicting late delivery risks and improving overall logistics performance.